WORKED SOLUTION

STEP II 1996, Question 4

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q4

The key idea

The square roots $\sqrt{1+x}$ and $\sqrt{1-x}$ are begging for $x = \cos t$, because the half-angle formulae turn $1 \pm \cos t$ into perfect squares. Once the roots are gone, the sum $\cos\theta + \sin\theta$ collapses into a single cosine, and the whole denominator becomes $1 + \cos(\text{something})$, which is again a perfect square. The final substitution recentres the interval so that the numerator becomes $\cos 4u$, which is exactly why the identity is proved first.

Solution

The identity

Using $\cos 2\theta = 2\cos^2\theta - 1$ twice, with $c = \cos u$:

$$\begin{aligned}\cos 4u &= 2\cos^2 2u - 1 = 2\left(2c^2-1\right)^2 - 1 \\ &= 2\left(4c^4 - 4c^2 + 1\right) - 1 = 8\cos^4 u - 8\cos^2 u + 1,\end{aligned}$$

as required.

The first substitution

Put $x = \cos t$, so $\mathrm{d}x = -\sin t\,\mathrm{d}t$. As x runs from -1 to 1, t runs from π down to 0; the minus sign flips the limits back, giving

$$I = \int_0^{\pi} \frac{\sin t}{\sqrt{1+\cos t} + \sqrt{1-\cos t} + 2}\,\mathrm{d}t.$$

Now $1 + \cos t = 2\cos^2\frac{t}{2}$ and $1 - \cos t = 2\sin^2\frac{t}{2}$.

Ask yourself Is $\sqrt{2\cos^2\frac{t}{2}}$ really $\sqrt{2}\cos\frac{t}{2}$, and not $\sqrt{2}\left|\cos\frac{t}{2}\right|$?

In general it is the modulus, but here $0 \le t \le \pi$, so $0 \le \frac{t}{2} \le \frac{\pi}{2}$ and both $\cos\frac{t}{2}$ and $\sin\frac{t}{2}$ are non-negative. This is precisely why the choice $t \in [0, \pi]$ (rather than, say, $[-\pi, 0]$) matters. Hence

$$\begin{aligned}\sqrt{1+\cos t}+\sqrt{1-\cos t} &= \sqrt{2}\left(\cos\tfrac{t}{2}+\sin\tfrac{t}{2}\right)\\ &= 2\left(\cos\tfrac{t}{2}\cos\tfrac{\pi}{4}+\sin\tfrac{t}{2}\sin\tfrac{\pi}{4}\right) = 2\cos\left(\tfrac{t}{2}-\tfrac{\pi}{4}\right).\end{aligned}$$

Writing $v = \frac{t}{4} - \frac{\pi}{8}$, so that $\frac{t}{2} - \frac{\pi}{4} = 2v$, the denominator is

$$2 + 2\cos 2v = 2(1+\cos 2v) = 4\cos^2 v = 4\cos^2\left(\tfrac{t}{4}-\tfrac{\pi}{8}\right),$$

and therefore

$$I = \int_0^{\pi} \frac{\sin t}{4\cos^2\left(\frac{t}{4}-\frac{\pi}{8}\right)}\,\mathrm{d}t,$$

as required.

The second substitution

Let $u = \frac{t}{4} - \frac{\pi}{8}$, so $t = 4u + \frac{\pi}{2}$ and $\mathrm{d}t = 4\,\mathrm{d}u$. The limits $t = 0$ and $t = \pi$ become $u = -\frac{\pi}{8}$ and $u = \frac{\pi}{8}$. The numerator is

$$\sin t = \sin\left(4u + \tfrac{\pi}{2}\right) = \cos 4u,$$

which is where the opening identity comes in. So

$$\begin{aligned}I &= \int_{-\pi/8}^{\pi/8} \frac{\cos 4u}{4\cos^2 u}\cdot 4\,\mathrm{d}u = \int_{-\pi/8}^{\pi/8} \frac{8\cos^4 u - 8\cos^2 u + 1}{\cos^2 u}\,\mathrm{d}u\\ &= \int_{-\pi/8}^{\pi/8} \left(8\cos^2 u - 8 + \sec^2 u\right)\mathrm{d}u.\end{aligned}$$

Dividing by $\cos^2 u$ is fine because $\cos u \ge \cos\frac{\pi}{8} > 0$ on the interval. Using $8\cos^2 u = 4 + 4\cos 2u$, the integrand is $4\cos 2u - 4 + \sec^2 u$, so

$$I = \Big[\,2\sin 2u - 4u + \tan u\,\Big]_{-\pi/8}^{\pi/8}.$$

The bracket is an odd function of u, so its value is twice its value at $\frac{\pi}{8}$:

$$\begin{aligned}I &= 2\left(2\sin\tfrac{\pi}{4} - \tfrac{\pi}{2} + \tan\tfrac{\pi}{8}\right) = 2\left(\sqrt{2} - \tfrac{\pi}{2} + \sqrt{2} - 1\right)\\ &= 4\sqrt{2} - \pi - 2,\end{aligned}$$

using $\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$ and the given $\tan\frac{\pi}{8} = \sqrt{2} - 1$. This is the required result.

Checks

- Numerically, $4\sqrt{2} - \pi - 2 \approx 0.51526$. Simpson's rule applied to the original integral in x, to the integral in t, and to the integral in u all give 0.515262.
- Size check: on $[-1, 1]$ the sum $\sqrt{1+x} + \sqrt{1-x}$ lies between $\sqrt{2}$ (at $x = \pm 1$) and 2 (at $x = 0$), so the integrand lies between $\frac{1}{4}$ and $\frac{1}{2+\sqrt{2}} \approx 0.293$. Hence $0.5 \leq I \leq 0.586$, and 0.515 fits.
- At $t = 0$ the denominator $4\cos^2(-\frac{\pi}{8})$ should equal $\sqrt{2} + 0 + 2$; indeed $4\cos^2\frac{\pi}{8} = 2(1 + \cos\frac{\pi}{4}) = 2 + \sqrt{2}$.

Common slips

- Forgetting the minus sign in $\mathrm{d}x = -\sin t\,\mathrm{d}t$, or forgetting to swap the limits, and so getting $-I$.
- Writing $\sqrt{1+\cos t} = \sqrt{2}\cos\frac{t}{2}$ without noting that $\cos\frac{t}{2} \geq 0$ on $[0, \pi]$.
- Dropping the factor 4 from $\mathrm{d}t = 4\,\mathrm{d}u$; it exactly cancels the 4 in the denominator.

Beyond STEP

Substitutions like $x = \cos t$ that "rationalise" square roots are a special case of a general fact: an integrand that is a rational function of x and $\sqrt{1-x^2}$ (or similar quadratic surds) can always be turned into a rational function by a trigonometric or the $t = \tan\frac{\theta}{2}$ substitution, and then integrated in elementary terms. At university you meet the systematic version of this, and also its limits: with cubic or quartic surds you get elliptic integrals, which are not elementary. For many more questions in this style, try Siklos, *Advanced Problems in Mathematics*.

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